Proposition 1 (Portmanteau). *The following are equivalent:*

i. $\mu_n \Rightarrow \mu$;

ii. $\lim_{n\to\infty} \mu_n(\phi) = \mu(\phi)$ *for all Lipschitz functions* ϕ;

iii. $\liminf_{n\to\infty} \mu_n(U) \geq \mu(U)$ *for all open sets* U;

iv. $\limsup_{n\to\infty} \mu_n(F) \leq \mu(F)$ *for all closed sets* F;

v. $\lim_{n\to\infty} \mu_n(A) \leq \mu(A)$ *for all continuity sets* A *of the measure* μ.

Lemma 2. *If* $f\colon I \to I'$ *is* C^3, *then* $f \prec 0$ *if, and only if,* f *increases cross ratios, in the sense that* $\rho(f(y_1), f(y_2), f(y_3), f(y_4)) > \rho(y_1, y_2, y_3, y_4)$ *for all* $y_1 < y_2 < y_3 < y_4$.

Lemma 3. *If* $f\colon I \to I$ *preserves orientation and* $f \prec 0$, *then* $f'(0)f'(1) < 1$.

Corollary 4. *If* $(x, y) \mapsto f_x(y)$ *is continuous and* $f_x \prec 0$ *for* μ*-almost every* $x \in X$, *then* $\mathrm{Lyap}(\mathcal{A}_0) + \mathrm{Lyap}(\mathcal{A}_1) < 0$. *In particular, at least one of the basins has positive measure.*

Lemma 5. *If* $(x, y_1) \in \mathcal{B}_0$, *then* $(x, y_2) \in \mathcal{B}_0$ *for all* $y_2 \leq y_1$. *Similarly, if* $(x, y_1) \in \mathcal{B}_1$, *then* $(x, y_2) \in \mathcal{B}_0$ *for all* $y_2 \geq y_1$.

Proof. We will prove the case $i = 0$, as the situation is symmetric. Let $(x, y_1) \in \mathcal{B}_0$ and $y_2 \leq y_1$. If $\varphi \in C(X \times I)$, and $\varepsilon > 0$, by the uniform continuity of φ, there exists $\delta > 0$ such that $y < \delta$ implies $|\varphi(x, 0) - \varphi(x, y)| < \varepsilon$ for all $x \in X$. Since $U = X \times [0, \delta)$ is a continuity set for ν_0 and $e_n(x, y_2) \to \nu_0$, by Portmanteau

$$\lim_{n\to\infty} e_n(x, y_2)(U) = \nu_0(U) = 1.$$

Define $\tau_n = e_n(x, y_2)(U) = \frac{1}{n}\#\{0 \leq i < n \mid f_x^n(y_2) < \delta\}$. Note that if $f_x^n(y_2) < \delta$, then $f_x^n(y_1) < \delta$ since the diffeomorphisms preserve orientation. Thus,

$$\begin{aligned} |e_n(x, y_1)(\varphi) - e_n(x, y_2)(\varphi)| &\leq \frac{1}{n}\sum_{i=0}^{n-1} \left|\varphi(g^n(x), f_x^n(y_1)) - \varphi(g^n(x), f_x^n(y_2))\right| \\ &\leq \varepsilon\tau_n + 2\|\varphi\|_\infty(1 - \tau_n). \end{aligned}$$

Since $\tau_n \to 1$, there exists N such that $n \geq N$ implies that the right side is less than 2ε, hence both integrals on the left converge to $\nu_0(\varphi)$. □

Theorem 6. *Suppose* $(x, y) \mapsto f_x(y)$ *is continuous and there is a metric* $d\colon I \times I \to \mathbb{R}$ *on* I *that satisfies:*

1. d *generates the usual topology on* I;
2. d *is uniformly stronger than the Euclidian metric, meaning that* $d(x_n, y_n) \to 0$ *implies* $|x_n - y_n| \to 0$;
3. *the map* F *is expanding for* d *in the interior of almost every fiber, meaning that for* μ*-almost every* $x \in X$ *and for all* $y_1, y_2 \in (0, 1)$ *with* $y_1 \neq y_2$, *we have* $d(f_x(y_1), f_x(y_2)) > d(y_1, y_2)$.

If at least one of the basins has positive Lebesgue measure, then there exists a measurable function $\sigma\colon X \to I$ *defined almost everywhere such that*

$$(x, y) \in \mathcal{B}_0 \text{ whenever } y < \sigma(x) \quad \text{and} \quad (x, y) \in \mathcal{B}_1 \text{ whenever } y > \sigma(x).$$

Consequently, the union $\mathcal{B}_0 \cup \mathcal{B}_1$ *has total* $\mu \otimes \mathrm{Leb}$ *measure.*

Proof. We start by defining functions $\xi_0, \xi_1 : X \to I$ that separate the basins in the following sense:

$$\xi_0(x) = \sup\{y \in I : y \in \mathcal{B}_0\}, \qquad \xi_1(x) = \inf\{y \in I : y \in \mathcal{B}_1\}.$$

By lemma 5, if $y < \xi_0(x)$ or $y > \xi_1(x)$, the empirical measures $e_n(x, y)$ converge to $\nu_0 = \mu \otimes \delta_0$ or to $\nu_1 = \mu \otimes \delta_1$, and if $\xi_0(x) < y < \xi_1(x)$, they do not converge to either ν_0 or ν_1.

It is sufficient to show that $A = \{x \in X : \xi_0(x) \neq \xi_1(x)\}$ has measure zero, since in this case we can define σ as the common value of the two functions. Suppose by contradiction that this set does not have measure zero. Since the f_x are bijective, A is an invariant set under g, so the ergodicity of μ implies that A has full measure.

From the assumption, we can assume without loss of generality that $\mathrm{Leb}(\mathcal{A}_1) > 0$, so by ergodicity, we also have $\xi_1(x) < 1$ for μ-almost every $x \in X$. Thus, given $k \in \mathbb{N}$, there exists $\delta_k < 1$ such that $E_k = \{x \in X : \xi_1(x) < \delta_k\}$ has measure $\mu(E_k) > 1 - 1/k$. Applying the Birkhoff ergodic theorem to the functions 1_{E_k}, we obtain $B \subseteq X$ of full measure such that for all $x \in B$ and k,

$$\lim_{n\to\infty} \frac{1}{n} \#\{0 \le i < n \mid \xi_1(g^i(x)) < \delta_k\} = \mu(E_k) > 1 - 1/k. \tag{1}$$

By the ergodicity of μ, there exists another set $C \subseteq X$ of full measure whose empirical measures converge to μ, and since f_x is expanding for d almost everywhere in $x \in X$, there exists a set D of full measure such that f_x is expanding for d for all $n \ge 0$ and $x \in D$. Let $x_0 \in A \cap B \cap C \cap D$ and $y_0 \in (\xi_0(x_0), \xi_1(x_0))$, and let $\nu \in \mathcal{M}(x_0, y_0) \setminus \{\nu_0\}$ be a limit point of the empirical measures different from ν_0. Note that since $x_0 \in B$, necessarily $(\pi_X)_*\nu = \mu$, but since $\nu \neq \nu_0$, it follows that $\nu(X \times \{0\}) < 1$. Thus, $\operatorname{supp} \nu$ is not contained in $X \times \{0\}$, and there exists $\varepsilon > 0$ such that $U = X \times (\varepsilon, 1]$ satisfies $\nu(U) > 0$. Let k be such that $\nu(U) > 1/k$, and $n_j \to \infty$ such that $\liminf_{j\to\infty} e_{n_j}(x_0, y_0)(U) > 1/k$. In other words, we have

$$\liminf_{j\to\infty} \frac{1}{n_j} \#\{0 \le i < n_j \mid f^i_{x_0}(y_0) > \varepsilon\} > 1/k. \tag{2}$$

Now, consider the sequence $r_n = d(f^n_{x_0}(y_0), f^n_{x_0}(\xi_1(x_0)))$, which is well-defined and strictly positive since $0 < f^n_{x_0}(y_0) < f^n_{x_0}(\xi_1(x_0)) < 1$ by the definition of the points. Note also that the graph of ξ_1 is strictly invariant under F, so $f^n_{x_0}(\xi_1(x_0)) = \xi_1(g^n(x_0))$.

Since $x_0 \in D$, it follows by lemma 2 that r_n is an increasing sequence. On one hand, it is not possible for $r_n \to \infty$, because from eqs. (1) and (2) we know that there exist $n \in \mathbb{N}$ arbitrarily large such that $0 < \varepsilon < f^n_{x_0}(y_0) < f^n_{x_0}(\xi_1(x_0)) < \delta_k < 1$. On the other hand, if $r_n \to r < \infty$, then for each $x \in X$ there exists a unique $\xi(x)$ such that $\rho(0, \xi(x), \xi_1(x), 1) = r$. In fact, a calculation shows that

$$\xi(x) = \frac{\xi_1(x)}{r(1 - \xi_1(x)) + \xi_1(x)}.$$

Note that $r_n \to r$ implies $\lim_{n\to\infty} |\xi(g^n(x_0)) - f^n_{x_0}(y_0)| = 0$. Since we know that x_0 is in the basin of μ, necessarily $e_n(x_0, y_0) \Rightarrow \eta$, where $\eta = (\operatorname{graph} \xi)_*\mu$ and $(\operatorname{graph} \xi)(x) = (x, \xi(x))$. Thus, η is an invariant measure. But then,

$$\varphi(x, y) = \frac{\rho(0, y, \xi_1(x), 1)}{1 + \rho(0, y, \xi_1(x), 1)}$$

is an integrable function defined at η-almost every point, with the property that $\varphi(F(x, y)) > \varphi(x, y)$ at η-almost every point. This is impossible, as it would imply $\int \varphi \, d\eta < \int \varphi \circ F \, d\eta = \int \varphi \, d\eta$. □